

AINOS Workflow Flowchart

Complete step-by-step operational flows — Registration to Revenue

- Auth ● Registration ● Dashboard ● Finance ● CRM ● Marketing ● AI Studio ● Inventory ● Billing
- Database / External

1

User Registration & Onboarding

New user signs up → company auto-created → dashboard access



Visit AINOS

yurekh.vercel.app/ainos





Choose Auth

Google OAuth or Email + Password





Fill Form

Name, Email, Password (min 6 chars)





POST /api/auth/register

Validate → bcrypt hash (12 rounds)

`route.ts`



Company Auto-Created

"Name's Company" in PostgreSQL

`prisma.company.create`



User Row Created

companyId assigned, role = "user"

`prisma.user.create`



Auto Sign-In

`signIn('credentials') → redirect /`

↓ Google OAuth Path ↓



Google Consent

User approves OAuth scopes



NextAuth Callback

/api/auth/[...nextauth]

Google profile → DB upsert



Find or Create

User + Company if first login

prisma upsert



Session Cookie

next-auth.session-token set



Dashboard

Sidebar + TopBar loaded

2

Invoice Creation Workflow (GST Billing)

Create invoice → auto GST calc → save → PDF → payment tracking



Dashboard

Sidebar → Finance → Invoices

/invoices



New Invoice

/invoices/new

page.tsx



Fetch Products

GET /api/products (companyId filtered)

+ GET /api/company



Select Customer

Name, GST, State, PAN



Add Items

Product, Qty, Price, HSN, GST%

↓ Auto-Calculation Engine ↓



Subtotal

$\Sigma (\text{qty} \times \text{price})$





Discount

Flat or % deduction



GST Split

Intra: CGST+SGST | Inter: IGST



Profit Calc

$\text{Total} - \text{costPrice} \times \text{qty}$



Grand Total

$\text{subtotal} - \text{discount} + \text{tax}$

↓ Save & Post ↓



POST /api/invoices

companyId + all fields

`prisma.invoice.create`





PostgreSQL

Invoice row with auto invoiceNumber



PDF Download

Template-based rendering with company branding



Payment Reminder

Auto-schedule if due_date set



Reports Update

Revenue + GST reports auto-refresh

3

AI Blog Agent — Generate & Publish Pipeline

Connect website → schedule → AI generates → publish to website → analytics



Connect Website

URL + niche + brand voice + tech stack

`/api/blog-agent/connect-website`



ConnectedWebsite

Saved with companyId

`prisma.connectedWebsite.create`





Create Schedule

Topic, keywords, tone, date, word count

`/api/blog-agent/schedule`



BlogSchedule

status = "pending"

`prisma.blogSchedule.create`

↓ AI Generation Triggered ↓



AI Writer

Gemini / Groq generates content

`/api/blog-posts/generate`



Featured Image

AI image generation

`Pollinations / MuAPI`



BlogPost Created

slug + content + image + SEO

`prisma.blogPost.create`



Schedule Updated

blogPostId linked, status = "generated"

`prisma.blogSchedule.update`

↓ Publish to Website ↓



Publish Trigger

Manual or auto on schedule date



embed.js Widget

Blog rendered via /blog/[slug] path routing on client website



WordPress

REST API push with app password credentials



Webhook

POST to custom webhook URL with blog JSON payload



Email Delivery

HTML email sent to deliveryEmail address

↓ Post-Publish ↓



Analytics Track

views, avgTimeOnPage, CTA clicks

`/api/blog-analytics`





Social Share

Auto-post to LinkedIn / Twitter if connected

ConnectedWebsite credentials



Conversion Track

Lead magnet clicks, form fills

/api/conversion-tracking

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AI Media Generation Studio

User prompt → credit check → muapi.ts → Pollinations/MuAPI → result URL



AI Studio

Sidebar → AI Studio → Media

/ai/media



Choose Mode

Image / Video / Image-to-Image / Cinema





Enter Prompt

Text description + settings (camera, lens, aperture)



Credit Check

GET /api/billing/credits

```
CreditBalance.balance ≥ cost?
```

↓ If credits sufficient ↓



POST /api/ai-media/generate

Prompt + model + params



muapi.ts Hub

Single integration layer

```
src/lib/muapi.ts
```



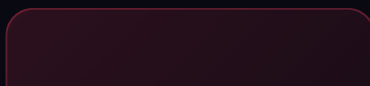
🌸 Pollinations.ai

Zero-cost image + video generation — instant URL, no polling

🎬 MuAPI.ai

Advanced models — paid, higher quality, more control

↓ Result ↓





Result URL

Image/Video URL returned to UI



Credit Deducted

CreditTransaction logged

`balanceAfter` recorded



ToolRun Saved

toolName, input, output, duration

`prisma.toolRun.create`



Download / Share

User saves or adds to Projects Gallery

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CRM Lead-to-Customer Pipeline

Lead captured → scored → followed up → converted to customer → deal closed



New Lead

Manual entry or auto-capture

POST /api/leads



Lead Row

status="new", score=0, companyld

prisma.lead.create



Score & Assign

Lead scoring + team assignment



Follow-Up

Create follow-up task with due date

POST /api/follow-ups

↓ Nurture ↓



Email Campaign

Drip sequence to lead

/api/email-campaigns



Contact Created

Promote lead → contact

POST /api/contacts



Deal Created

Value, stage, probability

POST /api/deals



Customer

Deal won → convert to customer

POST /api/customers

↓ Monetize ↓



Create Invoice

For the new customer

/invoices/new?customer=id



Payment Received

Mark invoice paid



Notification

Dashboard alert + TopBar bell

`prisma.notification.create`

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Inventory & Order Fulfillment

Product created → stock tracked → purchase order → sales order → delivery challan



Add Product

Name, SKU, HSN, GST%, MRP, cost

`POST /api/products`



Product + StockItem

Both created with companyId

`prisma.product.create`



Warehouse

Assign location + capacity

`POST /api/warehouses`



Stock Level

qty tracked, min/max alerts

↓ Low Stock? ↓



Purchase Order

Create PO for vendor

POST /api/purchase-orders



Vendor

Select/create vendor

POST /api/vendors



Receive Stock

PO fulfilled → stock incremented



Sales Order

Customer order → stock decremented

POST /api/sales-orders



Delivery Challan

GST-compliant delivery note

POST /api/delivery-challans

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Billing, Credits & Payment Flow

User buys credits → Razorpay payment → webhook → credit balance updated



Billing Page

Sidebar → Billing

/billing



Choose Plan

Select credit pack or subscription



Checkout

POST /api/billing/checkout

Razorpay order created



Razorpay Modal

User pays via UPI / Card / NetBanking

↓ Payment Success ↓



Razorpay Webhook

payment.captured event

`/api/webhooks/razorpay`



Verify Signature

HMAC SHA256 validation



CreditBalance Update

balance += credits purchased

`prisma.creditBalance.update`



CreditTransaction

type="credit", amount, balanceAfter

`prisma.creditTransaction.create`



Subscription Active

plan, status="active", period dates

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HR & Payroll Workflow

Employee added → attendance tracked → timesheet logged → payroll run → payslip



Add Employee

Name, dept, salary, bank

POST /api/employees



Daily Attendance

Check-in / Check-out

POST /api/attendance



Timesheet

Hours per project/task

POST /api/timesheets



Leave Request

Type, dates, reason → approval

POST /api/leaves

↓ End of Pay Period ↓



Payroll Run

Period, totalAmount, status

POST /api/payroll



Calculate

Salary – leaves + overtime



Approve & Process

status → "processed"



Notify Employee

Notification created

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Every API Request Lifecycle (Tenant Isolation)

How companyId isolation works on every single request



Browser

Dashboard UI action



Session Cookie

next-auth.session-token



NextAuth

Validate session → get user

`getServerSession()`



User.companyId

Extract tenant ID



API Route

Business logic runs



Prisma Query

WHERE companyId = ?

Tenant-scoped



JSON Response

Only THIS company's data

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Dashboard Navigation Map

All sidebar sections and their routes



Main

Dashboard /
Invoices /[invoices](#)
Customers /[customers](#)
Products /[products](#)
Calendar /[calendar](#)
Company /[company](#)
Services /[services](#)
Billing /[billing](#)
Reports /[reports](#)
Automations /[automations](#)
Settings /[settings](#)



Sales & CRM

Leads /[crm/leads](#)
Contacts /[crm/contacts](#)
Deals /[crm/deals](#)
Follow-ups /[crm/follow-ups](#)



Finance

Invoices /[invoices](#)
Expenses /[finance/expenses](#)
Quotations /[finance/quotes](#)
Challans /[finance/challans](#)
Tally /[finance/tally](#)



HR & Payroll

Employees /[hr/employees](#)
Recruitment /[hr/recruitment](#)
Attendance /[hr/attendance](#)
Timesheet /[hr/timesheet](#)
Payroll /[hr/payroll](#)
Leaves /[hr/leaves](#)

Reminders [/finance/reminders](#)

Templates [/finance/templates](#)

Inventory

Stock [/inventory/stock](#)

Warehouses [/inventory/warehouses](#)

Purchase Orders [/inventory/orders](#)

Sales Orders [/inventory/sales-orders](#)

Vendors [/inventory/vendors](#)

Marketing

Email Campaigns [/marketing/campaigns](#)

Social Media [/marketing/email](#)

SEO Platform [/marketing/seo](#)

Blog & Content [/marketing/blog](#)

Invitations [/marketing/invitations](#)

AI Studio

Chat [/ai/chat](#)

Media Studio [/ai/media](#)

Video Effects [/ai/effects](#)

Lip Sync [/ai/lipsync](#)

AI Skills [/ai/skills](#)

Projects Gallery [/ai/projects](#)

Support + More

Helpdesk [/support/helpdesk](#)

Knowledge Base [/support/knowledge-base](#)

Projects [/projects](#)

Compliance [/compliance/*](#)

Tools [/tools](#)